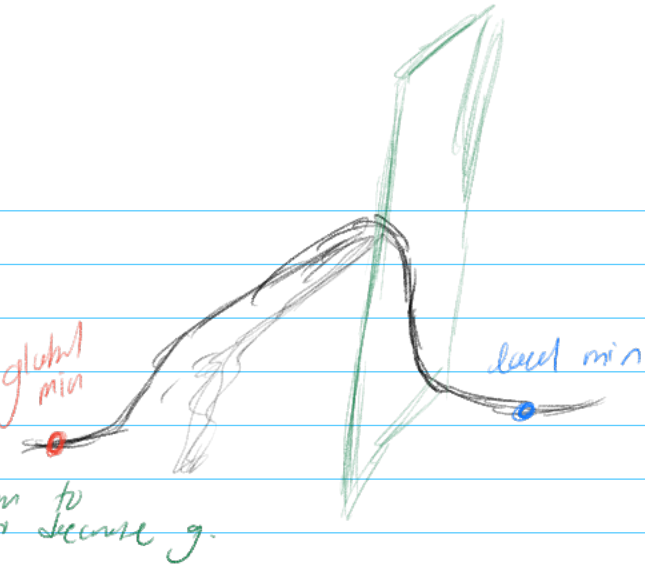
02/02/26

Steepest Descent Method

$$\vec{x}_k = \vec{x}_{k-1} - \alpha_{k-1} \nabla g(x_{k-1})$$

how much to descend

direction to move to decrease g.



solve using line search (in SDM)

choose α_{k-1} to "loosely" minimize on a line

$$\begin{aligned} f_1(x_1, x_2, \dots, x_n) &= 0 \\ f_2(x_1, x_2, \dots, x_n) &= 0 \\ &\vdots \\ f_n(x_1, x_2, \dots, x_n) &= 0 \end{aligned} \implies g(x_1, x_2, \dots, x_n) = \sum_{i=1}^{n} f_i^2(x_1, x_2, \dots, x_n)$$

root problem

min = 0 problem

(Quasi-) Newton's Method	Steepest Descent Method
• pro: converge fast	• pro: always converge to some local min
• con: need good initial guess	• con: converge slowly, not very accurate closer to min.

Numerical Linear Algebra

- Find $\vec{x}$ s.t. $A\vec{x} = \vec{b}$
- Finding $\lambda, \vec{v}$ s.t. $A\vec{v} = \lambda\vec{v}$

ex:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 4 \\ 3 & 5 & 7 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} \iff \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 4 & 4 & 3 \\ 3 & 5 & 7 & 5 \end{array}\right)$$

RREF

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{array}\right) \implies \begin{aligned} x_3 &= 1/2 \\ x_2 &= 0 \\ x_1 &= 1/2 \end{aligned}$$

row addition row operations $E_l + \lambda E_k \to E_l$ is equiv. to

$$A \to \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix} A$$

l row
k col

elementary matrices

$$A\vec{x} = \vec{b}$$

$$L_3 L_2 L_1 A = U \implies A = L_1^{-1} L_2^{-1} L_3^{-1} U \quad \text{let } L_1^{-1} L_2^{-1} L_3^{-1} = L$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} U = LU.$$

$\boxed{\text{LDU Factorization}}$ $A \in M_n(\mathbb{R})$ where $A = LDU$

L : lower $\triangle$
D : diag
U : upper $\triangle$

Start w/ $\tilde{L}\tilde{U}$ i.e. $A = \tilde{L}\tilde{U}$ & $D = \operatorname{diag}(\tilde{U})$
$U = D^{-1}\tilde{U}$
Then $A = \tilde{L}DU$

Ex:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 4 \\ 3 & 5 & 7 \end{pmatrix} \overset{LU}{=} \overset{\tilde{L}}{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{pmatrix}} \overset{\tilde{U}}{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix}}$$

let $D = \begin{pmatrix} 1 & & \\ 0 & 2 & 0 \\ 0 & & 2 \end{pmatrix}$ $\quad U = D^{-1}\tilde{U} = \begin{pmatrix} 1 & & \\ & 1/2 & \\ & & 1/2 \end{pmatrix}\tilde{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

$$\Rightarrow A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{U}$$

why is it unique?

LDL^T factorization for symmetric matrices $(A = A^T)$

$$A = LDL^T$$

$$A = \tilde{L}\tilde{U} = LDU \Rightarrow A^T = U^T D L^T$$

$$\Rightarrow LDU = U^T D L^T \qquad L = U^T \ \& \ U = L^T$$

$$\Rightarrow A = LDU = LDL^T$$

Cholesky Factorization

<u>def</u>: A matrix A is symmetric positive definite (SPD) if $A = BB^T$ for some non-singular matrix B ...